Knowledge Based Engineering (KBE)
AE4202

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Tutorial 2

Python vs. ParaPy object oriented programming

The ParaPy `classes`

- `Input()`, `@Input`
- `@Attribute`
-

`Class structure`

Agenda

- Python classes
- Para*Py* classes
- ParaPy GUI

Exercise 2 on Python classes
Exercise 3 on Para*Py* classes

Typical Python Class Layout

```
class MyClass(Superclass):                    # inherit from Superclass
    foo = 1                                     # class attribute

    def bar(self):                              # methods...
        return self.foo + 1

    def quz(self):
        return Box(width=self.bar)

    def qux(self, spam):
        print(spam)
```

indentation

extra arguments

Python classes

```
class Wing():
    taper = 0.2
    c_root = 4

    def get_c_tip(self):
        print("I take 10 minutes to return")
        return self.c_root * self.taper
```

Class variables

Class method

```
>>> wing = Wing()
>>> wing.get_c_tip()
"I take 10 minutes to return"
0.8
>>> wing.get_c_tip()
"I take 10 minutes to return"
0.8
>>> wing.taper = 0.3
>>> wing.get_c_tip()
"I take 10 minutes to return"
1.2
```

non-caching

Class initialization method

```
class Wing():

    def __init__(self, taper=0.2, c_root=4):
        self.taper = taper
        self.c_root = c_root
        self.c_tip = self.get_c_tip()

    def get_c_tip(self):
        print("I take 10 minutes to return")
        return self.c_root * self.taper
```

```
>>> wing = Wing()
"I take 10 minutes to return"
>>> wing.c_tip
0.8
>>> wing.c_tip
0.8
>>> wing.taper = 0.3
>>> wing.c_tip
0.8
>>> wing.c_tip = wing.get_c_tip()
"I take 10 minutes to return"
>>> wing.c_tip
1.2
```

non-lazy

caching

no dependency tracking

non-lazy

Typical Python Class Layout

```
class MyClass(Superclass):                    # inherit from Superclass
    foo = 1                                     # class attribute


    def bar(self):                              # methods...
        return self.foo + 1

    def quz(self):
        return Box(width=self.bar)

    def qux(self, spam):
        print(spam)
```

Typical ParaPy Class Layout

```
class MyClass(Base, Superclasses):        # inherit from ParaPy Base
    foo = Input(1)                        # Input

    @Attribute                            # Attribute
    def bar(self):
        return self.foo + 1

    @Part                                 # Part
    def quz(self):
        return Box(width=self.bar)

    def qux(self, spam):                  # method
        print(spam)
```

decorator

decorator

ParaPy adds dependency tracking, caching and lazy evaluation

Python Class

```
class Wing():
        taper = 0.2
        c_root = 4

        def get_c_tip(self):
            print("I take 10 minutes to return")
            return self.c_root * self.taper
```

```
>>> wing = Wing()
>>> wing.get_c_tip()
"I take 10 minutes to return"
0.8
>>> wing.get_c_tip()
"I take 10 minutes to return"
0.8
>>> wing.taper = 0.3
>>> wing.get_c_tip()
"I take 10 minutes to return"
1.2
```

ParaPy Class

```
class Wing(Base):
    taper = Input(0.2)
    c_root = Input(4)

    @Attribute
    def c_tip(self):
        print("I take 10 minutes to return")
        return self.c_root * self.taper
```

```
>>> wing = Wing()
>>> wing.c_tip
"I take 10 minutes to return"
0.8
>>> wing.c_tip
0.8
>>> wing.taper = 0.3
>>> wing.c_tip
"I take 10 minutes to return"
1.2
```

lazy

caching

lazy

dependency tracking

Test it yourself with `classes.py` on Brightspace

Input() and @Input: inputs to your ParaPy class object

```
class MyClass(Base):

    foo = Input()                    # required input

    foo = Input(<default>)           # optional input (simple)

    @Input                           # optional input (derived)
    def foo(self):
        return <expression>
```

no default

Use *only* if you actually have a good default!

different syntax, allows adaptive defaults (e.g. rule of thumb for sensible starting values)

Input examples

```
class Wing(Base):
      span = Input()
      AR = Input()

      c_root = Input(4.0)

     @Input()
     def c_tip(self):
         return self.span / self.AR / 2
```

This is *not* a good default… (see next slide)

`@Input:` expressions to compute default value, but can still be changed

Input defaults – what to use, and when?

- Benefits:
  - Users don't need to specify all inputs, and still get a complete Part
  - Things that rarely change don't need to be specified every time
- Dangers:
  - "Where does that number come from? I did not specify that!"
  - Accidentally treating input variables like constants
- Advice:
  - Provide defaults for values that rarely change: `n_fuselages = Input(1)`
  - Or for modifiable standard settings: `Mach = Input(0.1) # incompressible by default`
  - Defaults used for testing should be recognizably unrealistic!
    - E.g. default chord = thickness = span …. =1m
    - …but even better not to use defaults this way

Make *very obvious* that values are not *meant* to be realistic

Attribute: embodies engineering rule / is an output

```
⟶ @Attribute(**kwargs)
    def <name>(self):
  <expression>*
        return <expression>
```

```
class Wing(Base):
      # ...

      @Attribute
      def c_tip(self):
          return self.c_root *
  self.taper
```

See Exercise 3 code

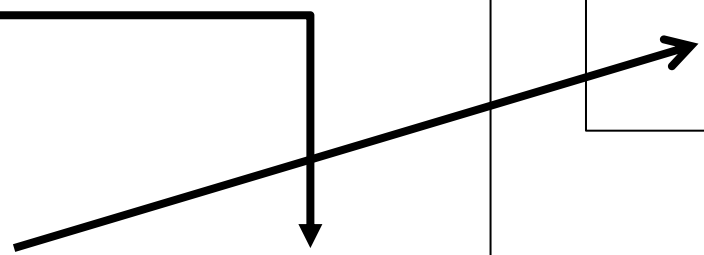
Part: composition of child objects

```
⟶ @Part(**kwargs)
    def <name>(self):
        return <Class>(*args,
                       hidden=bool,
                       suppress=bool,
                       quantify=int,
                       pass_down=str,
                       map_down=str,
                       **kwargs)
```

```
class Wing(Base):
    # ...

    @Part
    def airfoil:
        return
  Airfoil(chord=self.c_root)
```

```
class Airfoil(Base):
    chord = Input()
```



Single Part Example

```
class Wing(Base):
      thickness = Input(0.2)
      chord = Input(2)

      @Part
      def airfoil(self):
       return Airfoil(thickness=self.thickness,
                      chord=self.chord)
```

```
class Airfoil(Base):

    thickness = Input()
      chord = Input()
```

```
return Airfoil(pass_down=["thickness","chord"],
```

You can use `pass_down` for conciseness

```
>>> wing = Wing()
>>> wing.airfoil.thickness
0.2
>>> wing.airfoil.chord
2
>>> wing.airfoil.hidden
False
>>> wing.chord = 3
>>> wing.airfoil.hidden
True
>>> wing.chord = 1e-4
>>> wing.airfoil
Undefined
```

Quantified Part Example

```
class Airfoil(Base):
    thickness = Input(1)
    chord = Input(1)


class Wing(Base):
    n_airfoils = Input(2)
    thickness = Input(0.2)

    @Part
    def airfoils(self):
        return Airfoil(quantify=self.n_airfoils,
                       pass_down= =["thickness","chord"],
                       chord=0.1*(child.index+1),

     label="root_airfoil" if child.index == 0 else "other_airfoil")
```

```
>>> obj = Wing()
>>> obj.airfoils
<Sequence root.airfoils at 0x4b4f2e8>
>>> len(obj.airfoils)
2
>>> obj.airfoils[0].chord
0.1
>>> obj.airfoils[-1].chord
0.2
>>> [item.chord for item in obj.airfoils]
[0.1, 0.2]
```

See also exercise 3: How is airfoil chord set?

EXE: the geometry-less aircraft

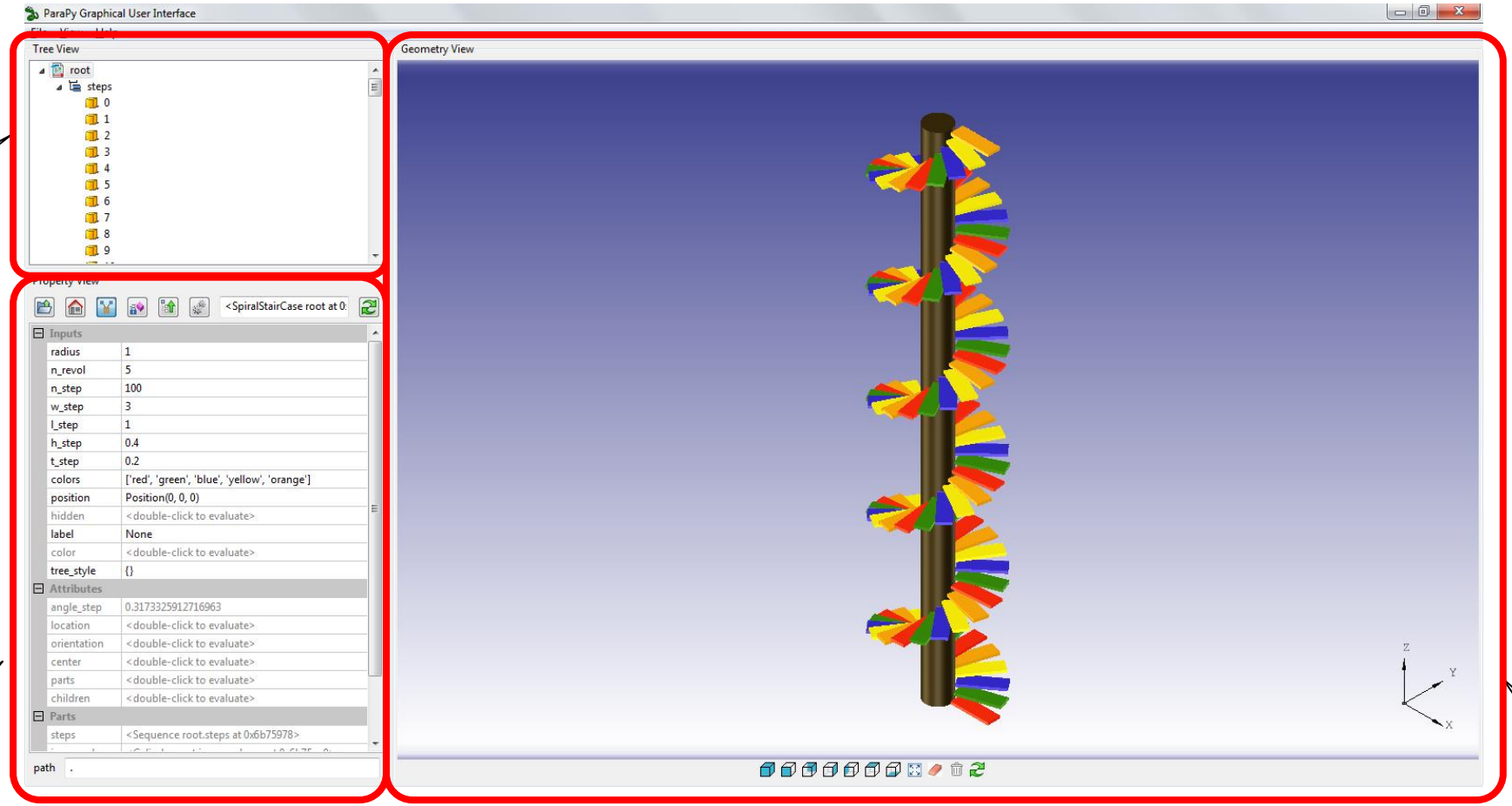
- Aircraft
- Fuselage
- Wing
- Engines & nacelles

EXE 4. See `Aircraft.py` on Brightspace

Observe:
- Class structure matches structure shown in parapy GUI
- How does the volume calculation work?

Add:
- Mass calculation (inspired by volume calculation)
- Some additional component, e.g.:
  - interiors in fuselage (use `quantify`)
  - High lift devices in wing
  - …

ParaPy Graphical User Interface (package: parapy.gui)

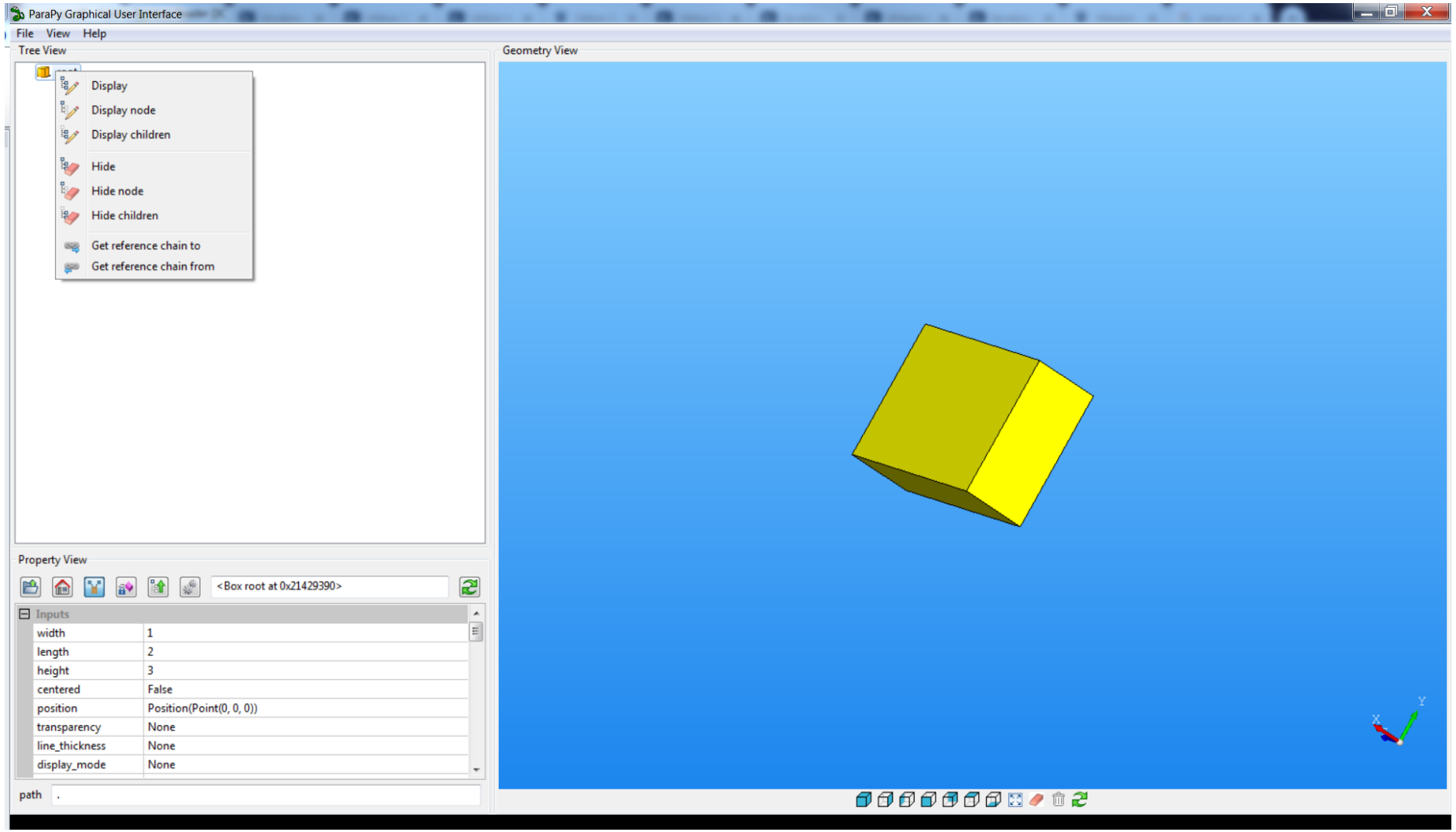


Product tree

Property Grid

Viewport

ParaPy GUI: Product Tree Context Menu

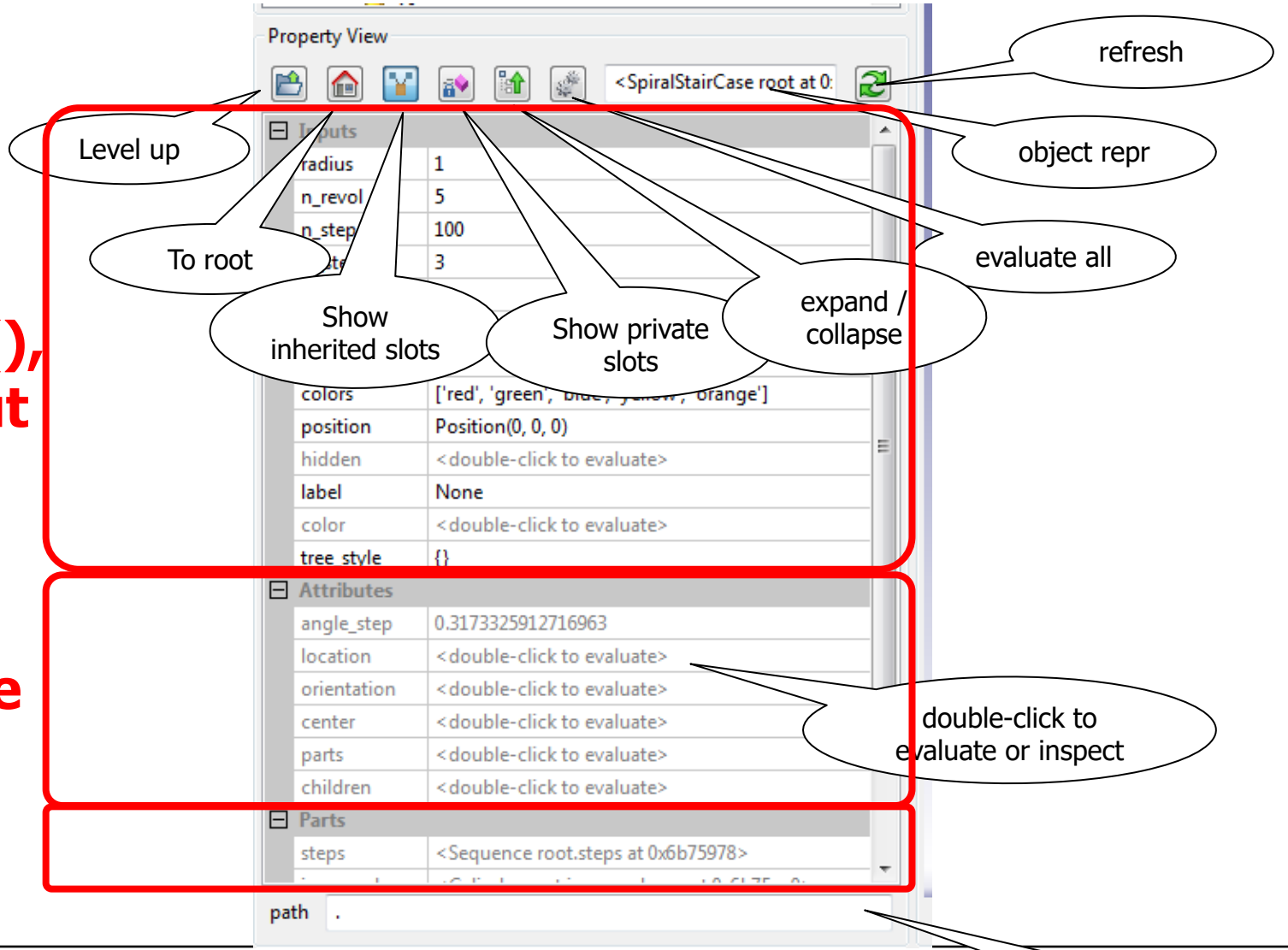


ParaPy GUI: Product Tree Context Menu

Product Tree

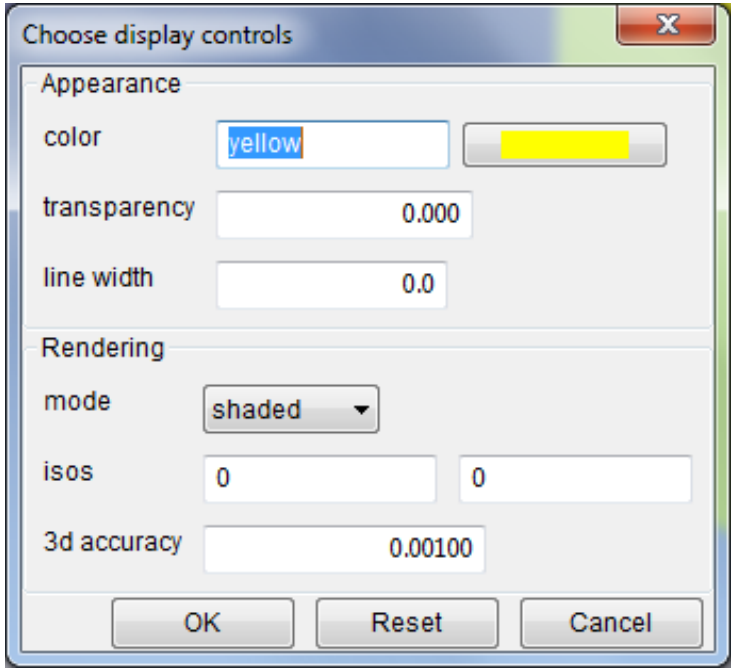
left-click	select node
double-click	display all
right-click	context menu
	draw this node and all children
	draw this node only
	draw children only
	hide this node and all children
	hide this node only
	hide children only
	reference chain to other object
	reference chain from other object
	bind object to 'self' (IPython only)

ParaPy GUI: Property Grid



ParaPy GUI: Viewport

f	fit all
c	hide all
w	wireframe
s	shaded
q	hidden-line removal
a	show tesselation
Delete	hide selected objects
left-click[1]	(un-)select object
left-click + motion	rotate
Shift + left-click[1]	(un-)select objects within area
right-click[1]	appearance dialog (with select)
right-click + motion[2]	dynamic zoom
Shift + right-click	area zoom
middle-scroll	zoom (x2)
Shift + middle-scroll	fine zoom (x1.2)
middle-click + motion	pan



[1] Hold Ctrl for incremental (un-)select. Object will be added to or removed from previous selection.

[2] Move the mouse horizontally.

ParaPy GUI: Viewport